

History of Integrated Circuits video presentation slides (2)

Condensed Study Notes

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LESSON ONE: INTEGRATED CIRCUIT (IC) Video Presentation

The development of Integrated Circuits (ICs) was driven by two main needs:

1. **Supporting Software Growth:** As software programs became more complex and sophisticated at a very rapid pace, the existing electronics industry needed to innovate. Without advancements in hardware technology to match the processing power demanded by new software, the progress of software development would be limited.
2. **Addressing Hardware Limitations:** Early mainframe computers were enormous and relied on vacuum tubes. These large components consumed a significant amount of electricity, making them inefficient and impractical. ICs offered a way to create much smaller and more power-efficient electronic circuits.

Think of it like this: Imagine building a super-fast race car (advanced software). If the roads (electronic hardware) are still made of dirt and cobblestones, the car can't reach its full potential. We needed better roads (ICs) to support the faster cars.

COMPUTER HARDWARE

Computer hardware refers to the physical components of a computer system. These are the parts you can see and touch, unlike the software that tells them what to do.

Think of computer hardware like the physical parts of a car: the engine, wheels, steering wheel, and dashboard. Software, on the other hand, is like the driver's knowledge and intentions – what makes the car move and go to a specific destination.

Integrated Circuits (ICs), which we discussed previously, are a prime example of crucial computer hardware. They are tiny electronic circuits built onto a small piece of semiconductor material, usually silicon. ICs allow for a massive reduction in the size and power consumption of electronic devices, making powerful computers possible.

3.0 Introduction to Integrated Circuits (ICs)

Integrated Circuits (ICs), also known by names like chips, microchips, or silicon chips, are a fundamental part of modern computer hardware. They are essentially a miniature city of electronic components, all built together on a tiny piece of semiconducting material, most commonly silicon.

Think of an IC like a highly organized, incredibly small neighborhood where all the houses (electronic components) are built right next to each other and connected by tiny roads (interconnections). The main 'residents' or building blocks within this neighborhood are transistors, but other components like diodes, capacitors, and resistors are also included.

From humble beginnings in the late 1950s, where early ICs had only about 10 components on a chip roughly 3mm square, the technology has advanced dramatically. The drive for more and more components on a single chip led to concepts like Very Large-Scale Integration (VLSI), which significantly increased the number of circuits that could fit. This advancement is what paved the way for the creation of the microprocessor.

To put the scale into perspective:

- The first commercially successful IC chip in 1974 contained around 4,800 transistors.
- By 1993, Intel's Pentium chip housed 3.2 million transistors.
- Today, it's possible to fit over a billion transistors onto a single chip, often no larger than a fingernail. This is a remarkable feat of 'cramming' a vast amount of electronic capability into an incredibly small space.

2.0 History of Integrated Circuits

The invention of the Integrated Circuit (IC) was a pivotal moment in technological history.

It was developed independently by two brilliant minds:

- **Jack Kilby** (1923-2005), who worked for Texas Instruments in the USA.
- **Robert Noyce** (1927-1990), who worked for Fairchild Semiconductor.

Although Kilby filed his patent first, Noyce's was granted earlier, leading to a legal discussion between their companies. Ultimately, in 1960, both companies agreed to share the idea, resulting in a shared patent.

Both Kilby and Noyce were recognized for their groundbreaking work, being inducted into the inventor's Hall of Fame.

The Integrated Circuit is widely considered the most significant and far-reaching technology developed in the 20th century, enabling the miniaturization and increased complexity of electronic devices we rely on today.

3.1 Fitting IC to a Circuit board

Integrated Circuits (ICs) are typically fitted onto circuit boards to be used in electronic systems. Figure 1.2 (as referenced in the material) shows a bread circuit board with an IC already inserted.

When working with ICs, it's important to orient them correctly. The 'Pin 1' of an IC is usually identified by a notch on the IC itself. This helps ensure the IC is placed in the correct orientation on the circuit board.

Connections to the IC are made via its pins. For example, a black wire, often used to denote a ground connection, is shown connected to 'GND' (Ground) on the circuit board. A red wire, often used to signify a power connection, is shown connected to a pin labeled 'Vcc' (which typically represents the power supply voltage).

Think of the circuit board as a miniature city and the IC as a special building. The pins are like the roads leading to and from this building. Pin 1's notch is like a specific landmark, such as a unique doorway, that tells you exactly which way the building faces so you can connect the roads (wires) correctly. 'GND' is like the city's main power outlet, and 'Vcc' is where you plug in the electricity to make the building work.

5.0 Classification of Integration Circuit

Integrated Circuits (ICs) can be classified in two main ways:

(1) Based on the type of transistors they use:

- **Bipolar Integrated Circuits:** These use bipolar junction transistors as their main component. They are generally preferred when the highest logic speed is needed.
- **MOS (Metal-Oxide-Semiconductor) Integrated Circuits:** These are used for larger-scale integration or when the lowest power dissipation is desired. High-performance bipolar and complementary MOS (CMOS) transistors can be combined (BiCMOS) to achieve both high speed and high density.

Both types of ICs rely on creating specific patterns of electrically active impurities within the semiconductor material and forming metal interconnections on its surface.

(2) Based on the scale of integration (the number of transistors or active circuit devices they contain):

- **Small-Scale Integration (SSI):** Contains fewer than 10 transistors.
- **Medium-Scale Integration:** Contains from 10 to 100 transistors.
- **Large-Scale Integration (LSI):** Contains from 100 to 1,000 transistors.
- **Very-Large-Scale Integration (VLSI):** Contains more than 1,000 transistors.

4.0 Types of IC

Integrated Circuits (ICs) can be broadly categorized into two main types based on how they process information: Analog ICs and Digital ICs.

1. Analog ICs (Linear ICs):

- These ICs work with signals that are **continuous** in nature. This means the signal can take on any value within a range, much like a dimmer switch for a light.
- Both the input and output signals are continuous.
- The output signal's level is directly related to (a function of) the input signal's level.
- Examples include Operational Amplifiers (OP Amps), Voltage Regulators, and Comparators.

2. Digital ICs:

- These ICs operate on **digital data**, which is represented in discrete values, typically as a logical '0' and a logical '1'.
- Think of digital signals like a standard light switch: it's either ON or OFF.
- In many digital systems, a logical '0' is represented by 0 Volts (0V), and a logical '1' is represented by a positive voltage, often around +5 Volts (+5V).
- Examples include Logic Gates, Counters, and Flip-flops.

6.0 Usage of IC

Integrated Circuits (ICs) are incredibly versatile building blocks in electronics, capable of performing a wide variety of functions.

ICs can be designed to act as:

- **Amplifiers:** These ICs increase the strength or amplitude of a signal. Think of them like a megaphone for electronic signals, making a weak signal louder.
- **Oscillators:** These ICs generate repetitive, oscillating signals, often used to create timing signals or carrier waves for communication.
- **Timers:** ICs that can be programmed to perform actions after a specific duration or to generate time-based pulses.
- **Counters:** These ICs keep track of events by counting pulses or signals they receive.
- **Memory:** ICs that store information, like the RAM (Random Access Memory) or ROM (Read-Only Memory) in a computer.
- **Microprocessors:** These are the 'brains' of a computer system. They are complex ICs that can execute a set of instructions to perform calculations and control other parts of the system.

7.0 Discrete Vs Integrated Circuit

Discrete circuits are built by connecting individual electronic components, such as Bipolar Junction Transistors (BJTs), Field Effect Transistors (FETs), diodes, resistors, and capacitors, to form a functional circuit.

Think of it like building with LEGOs. In a discrete circuit, you're using individual bricks (the components like transistors and resistors) and snapping them together one by one to create your structure (the circuit).

This is in contrast to Integrated Circuits (ICs), which we've learned are miniature assemblies of many components fabricated together on a single chip.

8.0 What is Doping Semiconductors

Semiconductors, like silicon, are the foundation of many electronic components, including Integrated Circuits (ICs). To make silicon useful for these circuits, we often modify its electrical properties through a process called **doping**.

Doping involves adding very small amounts of specific impurities to a pure semiconductor material, such as silicon. This deliberate addition changes how the material conducts electricity.

There are two main types of doping, leading to two types of modified silicon:

- **N-type Silicon:** This is created by doping silicon with an element like Phosphorus. Phosphorus has one more outer electron than silicon. When added to silicon, this extra electron is free to move, making electrons the predominant charge carriers.

- **P-type Silicon:** This is achieved by doping silicon with an element like Boron. Boron has one fewer outer electron than silicon. When added to silicon, this creates a "hole" where an electron should be. This hole can accept an electron, and it behaves like a positive charge carrier, making "holes" the predominant charge carriers.

Think of pure silicon as a perfectly organized dance floor. In N-type silicon, we've added a few extra dancers (electrons) who are eager to move around. In P-type silicon, we've created a few empty spots (holes) on the dance floor, and dancers from other spots are keen to fill them, effectively making the empty spot move.

8.1 P–N Junction

A p-n junction is the critical interface formed when two different types of semiconductor materials, p-type and n-type, are brought together within a single crystal.

Think of it like this: imagine a wall built between two neighborhoods. On one side (the p-type), there's an abundance of "holes" – think of these as empty parking spots available for electrons. On the other side (the n-type), there's an abundance of "electrons" – like cars looking for parking spots. These materials start out electrically neutral, meaning they have an equal number of positive and negative charges overall.

The key is that this junction is formed *inside* a single crystal, meaning the atoms are neatly arranged and connected. This intimate connection is what allows for interesting electrical behavior.

8.2 P-N Junction Diode

A P-N junction diode is created when a piece of N-type semiconductor material is joined with a piece of P-type semiconductor material.

Think of it like a one-way street for electricity. This junction is the fundamental building block for many electronic components.

(Figure 1.3 shows a P-N Junction diode.)

9.0 IC Fabrication Process

The manufacturing of Integrated Circuits (ICs) starts with a raw element like silicon. This silicon is then chemically treated, a process called **doping**, to give it specific electrical properties. Doping involves adding impurities to alter conductivity, creating either N-type (electron-rich) or P-type (hole-rich) materials, as we learned earlier.

Several methods are used for doping:

- **Sputtering:** This process shoots ions of the doping material at the silicon wafer, much like bullets from a gun.
- **Vapor Deposition:** Here, the doping material is introduced as a gas. It then condenses, forming a thin film of impurity atoms on the silicon wafer's surface.
- **Molecular Beam Epitaxy:** This is a more precise method of deposition.

Because ICs are built at microscopic or even nanoscopic scales, even a tiny speck of dirt can cause significant problems. To prevent this, ICs are manufactured in highly controlled environments called **clean rooms**. These rooms have meticulously filtered air, and workers must pass through airlocks while wearing specialized protective clothing.

Making ICs, which pack millions or billions of components onto a tiny silicon chip, is a complex process that involves more than just these initial steps.

a) The processes involved in IC fabrication

Fabricating Integrated Circuits (ICs) involves a series of carefully controlled steps, often repeated to build up the complex layers within an IC. Think of it like building a very intricate layered cake, but on a microscopic scale.

Here are the main processes:

1. **Making Wafers:** This is the very first step, where pure silicon crystals are grown into long cylinders. These cylinders are then sliced, much like slicing a salami, into thin, circular discs called wafers. Each wafer will eventually be divided into many individual IC chips.
2. **Masking:** The wafers are heated and coated with a protective layer of silicon dioxide. Then, a special light-sensitive material called **photoresist** is applied. Ultraviolet light is used to expose certain areas of the photoresist, creating a hard, protective pattern.
3. **Etching:** A chemical is used to selectively remove parts of the photoresist. This creates a stencil or **template pattern** on the wafer, revealing specific areas where we want to introduce changes, such as creating regions of n-type or p-type silicon.
4. **Doping:** The wafer, now with etched patterns, is heated in the presence of gases containing impurities. This process **dopes** specific areas of the silicon, transforming them into either n-type (electron-rich) or p-type (hole-rich) semiconductor materials, as we learned previously. Masking and etching steps might be repeated after doping to create more complex structures.
5. **Testing:** Once the fabrication steps are complete, each individual chip on the wafer is tested. A computer-controlled machine connects to the terminals of each chip. Any chip that doesn't function correctly is marked and discarded.
6. **Packaging:** The chips that pass the testing phase are carefully cut out from the wafer. They are then encased in protective materials, usually plastic, and fitted with external connection leads. This packaging makes them ready to be used in electronic devices like computers.

9.1 A summary of IC Fabrication Process

The creation of Integrated Circuits (ICs) on silicon wafers involves a multi-step process. While the exact details are complex, the overall journey from raw silicon to a functional chip is a remarkable feat of engineering.

Think of IC fabrication like building a miniature city on a silicon "foundation." Each step adds or modifies tiny structures that will eventually form the electronic pathways and components of the IC.

We've previously touched on the initial steps like growing the silicon and doping it to create specific electrical properties. The subsequent steps build upon this foundation to form the complex circuits within the IC.

b) Quality Control

Even with highly controlled environments and precise tools during Integrated Circuit (IC) fabrication, a significant number of chips are rejected. While the percentage of rejected chips has decreased over time, creating intricate networks of microscopic circuits remains a challenging process, making some level of rejection unavoidable.

Think of it like baking a very complex cake with many delicate layers. Even with the best ingredients and careful instructions, sometimes a layer might break, or a decoration might not turn out perfectly, leading to that cake not being perfect enough to sell.

c) Hazardous Materials and Recycling

The fabrication of Integrated Circuits (ICs) involves the use of certain dopants, such as gallium and arsenic. These substances are classified as toxic, meaning their handling, storage, and disposal require strict control to ensure safety and environmental protection.

Due to the widespread usefulness and versatility of ICs, a robust recycling industry has emerged. This industry involves salvaging ICs and other electronic components from discarded or outdated equipment. These salvaged components are then tested to ensure they are still functional and subsequently resold for integration into new electronic devices. This practice helps to reduce waste and extend the lifespan of valuable electronic parts.